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Editors
Transportation Research Part E: Logistics and Transportation Review

Dear Editors,

Please consider the enclosed manuscript, "Cost, Carbon, and the Speed of Replenishment in Distribution Network Design," for publication in *Transportation Research Part E: Logistics and Transportation Review*.

Firms continue to revisit how many distribution centers (DCs) to operate and how fast to replenish them, mainly for cost and delivery service; carbon pricing and emissions reporting now bear on them as well. Slower line haul burns less fuel but requires more safety stock and stock in transit; more DCs shorten the last mile but add facilities and give up the pooling that keeps inventory low. We model network density and replenishment speed jointly, under a carbon cap and a service requirement.

We establish that the joint design has a unique optimum at every carbon price and that speed and density are complements. The density threshold that separates the cost- and carbon-minimizing designs accounts for lost pooling and for the carbon-minimizing firm's speed, so inbound freight carbon enters the choice of how many DCs to operate through safety stock.

We also characterize how a tighter carbon cap changes the least-cost design. For a given network, it favors slower replenishment when faster freight emits more per dollar than the inventory it saves, as when speed is bought with fuel, and faster replenishment in the reverse case. For a given speed, it favors a denser network when last-mile delivery emits more per dollar than the additional facilities and safety stock. When both decisions adjust, the initial response can differ from the eventual design: a modest carbon target can favor consolidation even when the carbon-minimizing network is more dispersed. A calibration to public US data illustrates the least-cost way to meet a carbon target.

The manuscript fits the journal's focus on logistics and operations models with application and on the evaluation of public policies. It is original, has not been published, and is not under consideration elsewhere.

Sincerely,

Ram Ganeshan